

Cardiac Field Artifact Removal in Ear-EEG Using Repetitive Disturbance Rejection

Wenlin Zhang¹, Sam Lin^{1,6}, Shuai Li³, Ye Yuan¹,
Zheng Wang^{1,2}, Ziteng Ni⁴, Heng Kang Yao^{2,4}, Yongcheng Li⁵, Davide Momi⁷

Abstract—Continuous wearable EEG is increasingly paired with consumer wearables that provide a heartbeat stream (e.g., smartwatch ECG/PPG). Taking the available ECG/heartbeat reference, we propose a repetitive-control framework that models cardiac artifact as a heartbeat-aligned periodic disturbance. Each EEG sample is assigned a cardiac phase index wrapped around ECG R-peaks; phase-synchronous averaging then preserves heartbeat-correlated residue while attenuating non-phase-locked noise. Subtracting this phase-indexed average yields effective artifact removal in the non-causal offline setting. For online real-time operation, an iterative-learning estimator is used to track the residue in either a same-beat predictive mode (strict causal) or a one-beat-delayed mode (buffered causal). The effectiveness of proposed methods is demonstrated through both real and semi-synthetic data experiments.

I. INTRODUCTION

Wearable EEG is moving toward continuous, day-scale monitoring, and many users already wear a second device that provides heartbeat timing from ECG or PPG. This creates a practical path for low-channel ear-EEG denoising: use heartbeat timing already available from the wearable ecosystem rather than requiring additional EEG channels. In ear-centered montages, cardiac field artifact (CFA) is often prominent because electrodes are close to neck and vascular pathways. CFA is the volume-conducted electrical field of the heart [1], distinct from the scanner-related ballistocardiogram artifact emphasized in EEG-fMRI work [2]. For out-of-scanner earables, CFA is therefore a primary contamination mechanism.

Classical ECG-referenced adaptive cancellation is well established [3], and it is a natural baseline. However, in our target regime it faces two practical limits: earables are channel-starved for blind source separation [6], and reference-to-artifact waveform predictability can be weak under real volume-conduction mismatch and nonstationarity. Our real-data observations in this paper support exactly this point: coherence between ECG and ear-EEG artifact is diluted with beat-synchronous pooling, so reference-driven waveform regression is limited as a primary mechanism. At the same time, beat timing remains highly informative, because the artifact is still phase-locked to R-peaks.

Affiliations: ¹Institute of Brain and Mind, CA, USA; ²Krembil Centre for Neuroinformatics, Centre for Addiction and Mental Health, Toronto, Ontario, Canada; ³University of Oulu, Oulu, Finland; ⁴Synchroni Scientific Instruments Inc., Canada; ⁵Shenzhen Institutes of Advanced Technology, China; ⁶Palos Verdes Peninsula High School, CA, USA; ⁷University of California, Irvine, CA, USA; Corresponding author: Wenlin Zhang (e-mail: william@ibmind.org).

This motivates a timing-only control view: model cardiac contamination as a repetitive disturbance and reject it by heartbeat phase indexing. Repetitive control and iterative learning control (ILC) are mature tools for periodic disturbance rejection in mechatronics [8], and they map naturally to the heartbeat cycle when reliable beat timing is available. In this work, we focus on phase-aligned template learning/cancellation driven by heartbeat timing in three deployment modes: non-causal offline, buffered causal (one-beat latency), and strict real-time causal (zero added latency).

The resulting contribution is a compact, wearable-oriented CFA removal scheme aligned with realistic device constraints: low channel count, low compute budget, and timing references that may come from a separate wearable.

The rest of this paper is organized as follows. Section II states the signal model and timing-only CFA-removal problem. Section III develops repetitive template cancellation in non-causal and causal forms. Section IV-A reports experiments on real ear-EEG and semi-synthetic data. Section V summarizes the main findings and practical implications.

II. PROBLEM FORMULATION

Following repetitive-control modeling conventions [8], [9], we model the measured ear-EEG signal as

$$y(t) = s(t) + d_r(t) + n(t), \quad (1)$$

where $s(t)$ is the neural signal of interest, $d_r(t)$ is the heartbeat-synchronous repetitive cardiac disturbance, and $n(t)$ aggregates asynchronous interference and sensor noise. The subscript r denotes the repetitive nature of the disturbance. Its period is heartbeat dependent and is set by ECG R-peak timing, with beat interval

$$T_k = t_{k+1} - t_k, \quad (2)$$

where t_k is the k -th detected R-peak instant.

Assumption 1. *In ear-EEG, CFA is dominated by the repetitive component $d_r(t)$. The R-peak phase-asynchronous disturbances and noise $n(t)$ are assumed zero-mean across the R-peak sync averaging, and DC offset and slow drift are removed by preprocessing.*

Assumption 2. *The neural signal of interest $s(t)$ is not phase-locked to the R-peak: across beats, its phase relative to the cardiac cycle is effectively uniform. Consequently $s(t)$ is preserved by the proposed phase-synchronous repetitive runout cancellation, more detailed analysis is provided in Sec. III.*

Using heartbeat phase, we represent the repetitive disturbance as a phase-indexed waveform

$$d_r(t) = d_r(\phi(t)), \quad \phi(t) = \frac{t - t_k}{T_k}, \quad t \in [t_k, t_{k+1}), \quad (3)$$

with $\phi(t) \in [0, 1)$ the R-peak-aligned cardiac phase.

Assumption 3. *The beat period T_k is assumed to be accurately locked, and slowly varying due to respiratory sinus arrhythmia (RSA), with additional small modulation from low-level muscle/motion effects in resting-state recordings.*

Problem 1. *Find the repetitive cardiac artifact template $\hat{d}_r(\phi(t))$ such that subtracting it from the measured signal recovers the neural signal of interest:*

$$\hat{s}(t) = y(t) - \hat{d}_r(\phi(t)) \approx s(t). \quad (4)$$

Under Assumptions 1–3, $\hat{d}_r(\phi(t))$ can be estimated from the R-peak-locked structure of $y(t)$ alone.

Note 1. *Since $d_r(\phi(t))$ is periodic in the cardiac phase $\phi(t)$ indexed solely by R-peak instants $\{t_k\}$, Problem 1 depends only on R-peak timing and never on the ECG waveform shape. Consequently, any wearable heartbeat source—smartwatch ECG, wrist PPG, or any device delivering beat timing via a synchronisation mechanism—is sufficient to instantiate the solution.*

Note 2. *In reality, $s(t)$ may contain a phase-synchronous part that has non-zero expectation mean value across beats, e.g. the heartbeat-evoked potential (HEP)—a genuine cortical response time-locked to the cardiac cycle may or may not be absorbed into $\hat{d}_r$. In this paper, we make no claim that the solution of Eq. 4 preserves the phase-synchronous part of the neural signals, and it's a meaningful future work or extension worthy of further investigation.*

To instantiate Problem 1 on the discrete sample grid, let $\hat{t}_k$ be the estimated k -th R-peak sample instant and f_s the sampling rate. The k -th beat's R-peak-aligned $d_r(\phi(t))$ at integer offset p is defined as

$$d_k\left(\hat{t}_k + \frac{p}{f_s}\right) \triangleq d_k(\phi_p), \quad (5)$$

with the discrete phase index and offset range

$$\phi_p \triangleq \phi\left(\hat{t}_k + \frac{p}{f_s}\right) = \frac{p}{f_s T_k}, \quad (6a)$$

$$p \in \{-n_{\text{pre}}, \dots, -1, 0, 1, \dots, n_{\text{post}}\}. \quad (6b)$$

Note 3. *The template window spans $n_t = n_{\text{pre}} + n_{\text{post}} + 1$ samples ($p = 0$ at the R-peak). A practical choice is $n_{\text{pre}} = \lfloor 0.05 f_s \rfloor$ (50 ms, covering QRS onset) and $n_{\text{post}} = \lfloor 0.55 f_s \rfloor$ (550 ms, covering the full QRS-T complex); at $f_s = 250$ Hz this gives $n_t = 151$ samples.*

III. REPETITIVE DISTURBANCE REJECTION

A. Non-Causal Cancellation

For offline CFA removal, we could run non-causal averaging to obtain the solution of Problem 1. That is, for a fixed phase

index p (offset from the R-peak), the template value is the average, across N_r beats, of the EEG sampled at the *same* offset p from each detected R-peak $\hat{t}_k$:

$$\hat{d}_r(p) = \frac{1}{N_r} \sum_{k=1}^{N_r} y_k(p), \quad y_k(p) \equiv y(\hat{t}_k + p/f_s). \quad (7)$$

Substituting the signal $y_k(p)$ by its definition from the discrete time version of Eq. (1), we obtain

$$\hat{d}_r(p) = \underbrace{d_r(p)}_{\text{same every beat}} + \frac{1}{N_r} \sum_{k=1}^{N_r} s_k(p) + \frac{1}{N_r} \sum_{k=1}^{N_r} n_k(p). \quad (8)$$

In Eq. (7), we use $\hat{d}_r(p)$ to denote $\hat{d}_r(\phi_p)$ for better readability, and so are the p -indexed notations used in the subsequent discussion.

Take the expectation of Eq. (8) over R-peaks, we obtain

$$\mathbb{E}[\hat{d}_r(p)] = d_r(p) + \mathbb{E}[s_k(p)] + \mathbb{E}[n_k(p)]. \quad (9)$$

The repetitive R-peak-locked disturbance template $d(p)$ is deterministic. Ideally, The non-R-peak-locked neural term $\mathbb{E}[s_k(p)]$, and noise term $\mathbb{E}[n_k(p)]$ will render to zero according to Assumption 1, therefore the Eq. (8) will offer us solution by design:

$$\hat{d}_r(p) \approx d_r(p), \quad (10)$$

The CFA removed EEG signal is approximated by

$$\hat{s}_k(p) \approx y_k(p) - \hat{d}_r(p), \quad (11)$$

Let the asynchronous part at offset p have per-sample variance

$$\text{Var}[s_k(p) + n_k(p)] = V(p). \quad (12)$$

The noise floor leakage in $\hat{d}_r(p)$ due to limited sample average is the sample-mean of these N_r terms. If the per-beat samples are *mutually uncorrelated*,

$$\text{Var}[\hat{d}_r(p)] = \frac{1}{N_r^2} \sum_{k=1}^{N_r} V(p) \approx \frac{V(p)}{N_r} (1 + \delta_{coh}), \quad (13)$$

If we consider realistically there are some neuro signal has slight correlation with the R-peaks, the leakage will have an inflation term as in Eq. (13), where δ_{coh} represents the total correlation accumulation across the beats. However, as the we choose a reasonable large enough N_r , the leakage and inflation will becomes negligible, we will give more detailed analysis in the next sections.

B. Causal Cancellation

For streaming operation the batch average of Eq. (7) is replaced by its standard exponentially-weighted ILC recursion, updated once per learnable beat at each offset p :

$$\hat{d}_r^{(k)}(p) = \hat{d}_r^{(k-1)}(p) + a [y_k(p) - \hat{d}_r^{(k-1)}(p)], \quad (14a)$$

$$\hat{s}_{k-1}(p) \approx y_{k-1}(p) - \hat{d}_r^{(k-1)}(p). \quad (14b)$$

where a is a tunable learning gain selected from the convergence-versus-leakage tradeoff; the supervisor freezes

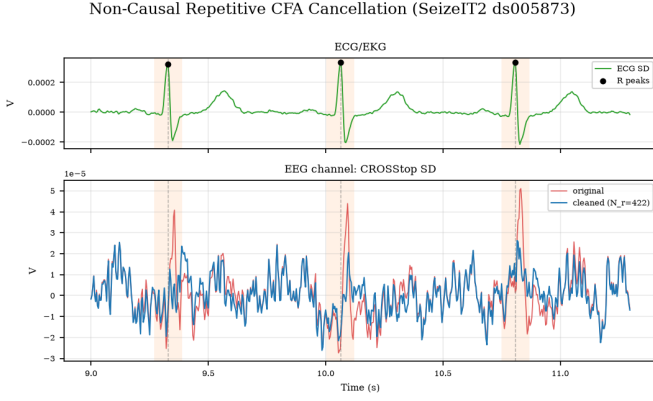


Fig. 1. Non-causal repetitive CFA cancellation on a short SeizeIT2 segment (ds005873, sub-070, ses-01, run-20; EEG index. Top: chest ECG with detected R-peaks (green). Bottom: original (red) and non-causally cleaned (blue) ear-EEG. The repetitive template $\hat{d}_r(p)$ by (Eq. (11)) is estimated from the full recording session ($N_r > 800$ beats).

the update on premature/missed beats. Eq. (14b) is the buffered one-R-peak-delay cancellation form, equivalent to Eq. (11) with index $k - 1$. For true strict real-time cancellation, the current beat's R-peak t_k has not arrived yet, so the canceller must anchor phase using the PI-type R-peak phase tracker prediction rather than detected t_k timing; this enables zero added latency at the cost of prediction-jitter-limited suppression around the R spike. Both modes are $O(1)$ per sample with $O(L_w)$ memory, which makes the proposed method favorable for real-time implementation.

Note 4. *Real-time cancellation requires that the R-peak detector has basic capability to reject immature beats, or to invalidate abnormal R-peaks, so that both template update and cancellation can be held when the timing reference is unreliable. We provide a functional robust R-peak detector and phase tracker for data analysis in the next subsection; however, detailed detector/tracker design is outside the scope of this paper.*

IV. EXPERIMENTS

For implementing and evaluating the proposed method, we developed a functional robust R-peak detector and PI phase tracker for both real-data and simulation-data analysis [12]. Especially for the real-data showcase, We only demonstrate the RRO-template canceller on records where reliable phase locking is available by our detection algorithm, and defer general-purpose phase tracking to future work.

A. Real Ear-EEG Validation

In this subsection, we mainly use the two-channel ear-EEG configuration for discussion and analysis from the SeizeIT2 dataset. SeizeIT2 is a multicenter wearable epilepsy dataset containing long-duration multimodal recordings (behind-the-ear EEG, ECG, EMG, ACC, and GYR) with focal-seizure annotations across a large patient cohort [13].

Fig. 1 shows the result of non-causal repetitive disturbance rejection to the SeizeIT2 segment: the template is average

across $N_r = 851$ learnable beats, and the qualitative view of this figure shows the repetitive structure of the CFA removal by the proposed method.

Template SNR via a phase-scramble null. Since the true disturbance d_r is unobservable on real recordings, we define template quality through a phase-scramble null SNR rather than a direct error to d_r . The ground-truth disturbance d_r is unobservable, so we validate the template against a null that shares its statistics but destroys R-phase locking. Define the R-locked energy of a signal z under a peak set P ,

$$\mathcal{E}_P[z] = \left\| \frac{1}{|P|} \sum_{k \in P} z(t_k + p/f_s) \right\|_2^2, \quad (15)$$

summed over the template window. Averaging on the true peaks gives $\hat{d}_r = d_r + \ell$, with leakage ℓ of scale $SD/\sqrt{N_r}$ (Eq. (13)); a circular time-shift of the peak train, which preserves the RR-interval sequence but destroys absolute phase, gives leakage only. Their ratio,

$$Q = \frac{\mathcal{E}_{\text{true}}[y]}{\mathcal{E}_{\text{scr}}[y]}, \quad \text{SNR}_{\text{templ}} = 10 \log_{10} Q, \quad (16)$$

is the CFA energy standing above the RRO-removed noise floor. Furthermore, we define the bias-corrected CFA RMS as the effectiveness of CFA removal:

$$A/SD = \frac{\sqrt{\max(\mathcal{E}_{\text{true}} - \overline{\mathcal{E}_{\text{scr}}}, 0)/L_{\text{lobe}}}}{\text{std}(y)}. \quad (17)$$

Here L_{lobe} is the number of samples in the main-lobe window around the R-peak (default window: 50 ms pre and 150 ms post), and $SD = \text{std}(y)$ is computed over the full preprocessed EEG segment used for that run. For the real-data example in Fig. 1, the metric reads $\text{SNR} = 28.80$ dB and $A/SD = 1.061$, indicating a strong and clearly resolvable R-locked component. From the whole dataset, we could observe above 20dB $\text{SNR}_{\text{templ}}$ spanning 319 runs from 36 subjects, with $A/SD \in [0.110, 1.220]$ and mean $A/SD = 0.557$, as a showcase of the proposed method's effectiveness. Again, robust R-peak/beat-phase estimation on arrhythmic or low-quality ECG, and systematic characterization across the full dataset, are outside the scope of this work;

Coherence view. Template fidelity does not imply that the artifact waveform is predictable from the ECG. Fig. 2 contrasts the same R-locked ECG window against three EEG views: the raw ear-EEG beat, the stacked non-causal template $\hat{d}_r$, and the buffered one-beat-delay template. This separates two distinct properties. First, the left panel shows that the cardiac artifact is clearly timing-repeatable: both templates lock to the same R-centered structure. Second, the right panel shows that waveform predictability from the ECG remains weak even after R-locking, with the raw EEG coherence especially diluted by the neural background and the template coherences still far from a near-deterministic waveform map. That is precisely the regime in which a timing-only canceller is the appropriate tool and reference-driven waveform regression has little coherent structure to act on.

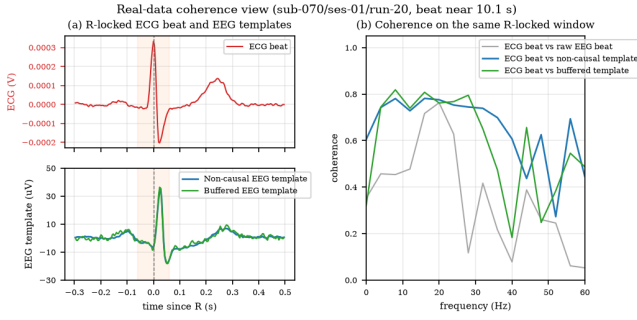


Fig. 2. Coherence view on the same real recording. Left: one R-locked ECG beat overlaid with the non-causal EEG template and the buffered one-beat-delay EEG template. Right: coherence between the windowed ECG beat and three R-locked EEG views: the raw EEG beat, the stacked non-causal template, and the buffered template. The templates preserve clear heartbeat timing structure, but the ECG-to-EEG waveform coherence remains weak, which is exactly the regime in which timing-only cancellation is useful and direct waveform regression is limited.

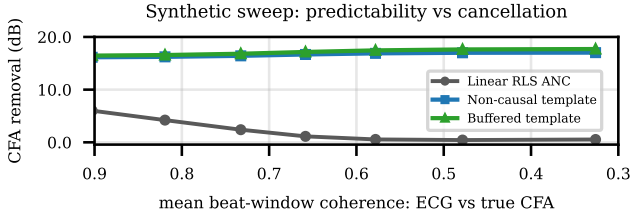


Fig. 3. Synthetic sweep of reference predictability versus cancellation performance. The x-axis is the mean 1–20 Hz coherence between the averaged ECG beat and the averaged *true* synthetic CFA beat under a controlled sweep that blends an ECG-coherent linear artifact with a phase-locked but waveform-decorrelated component at fixed artifact power. The y-axis is SNR improvement over the contaminated input, computed against the true neural signal. Standard linear RLS adaptive noise cancellation improves as coherence rises and degrades as coherence falls, while the non-causal and buffered repetitive template methods remain effective because they rely on timing repeatability rather than waveform predictability.

B. Synthetic Simulation

In this subsection, we turn to a semi-synthetic simulation in which the ECG-to-artifact channel is swept from linear (high coherence) to strongly nonlinear (near-zero coherence) at fixed artifact power. The popular ICA-based CFA removal can be effective with sufficiently many channels, but not in low channel count wearable settings. Thus, we only compare with standard RLS type of reference based cancellation method. Fig. 3 clarifies the practical regime of low-channel ear-EEG, by comparing a linear ECG-referenced baseline (RLS-ANC) against the proposed repetitive disturbance rejection under controlled predictability change. RLS-ANC behaves as a coherence-gated linear canceller and it only suppress the CFA to the noise floor, our synthetic simulation has to choose a higher SNR CFA simulation than the typical real-data scenario. By contrast, the non-causal and buffered repetitive methods remain comparatively stable because they estimate the heartbeat-locked disturbance directly from EEG using beat timing, thereby removing not only the linearly coherent

part but also nonlinear and coherence-diluted heartbeat-locked components embedded in neural activity and measurement noise.

V. CONCLUSION

We have presented a control-theoretic view of cardiac-field-artifact (CFA) removal in low-channel ear-EEG: CFA is modeled as a heartbeat-phase-locked repetitive disturbance and rejected by phase-indexed synchronous averaging and iterative learning, anchored only to R-peak timing and never to the ECG waveform. We applied the proposed repetitive template method to both real datasets and synthetic simulation, demonstrating its effectiveness in removing cardiac-field artifacts with only R-peak timing as a reference phase index.

More broadly, this work aims to bridge mature mechatronic disturbance-rejection theory and biosignal denoising, opening a control-systems avenue for wearable neural interfaces. This perspective extends naturally beyond the cardiac artifact: an inertial (gyroscope/accelerometer) reference could drive feed-forward cancellation of motion artifact by the available sensors.

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